In this case we say that the operator $\mathcal{L}_f$ is *skew-adjoint*. The equations

$$\frac{\partial g}{\partial t} = \mathcal{L}_f g,$$

for the solution operator, and

$$\frac{\partial \rho}{\partial t} = \mathcal{L}_f^\dagger \rho = -\mathcal{L}_f \rho,$$

for the propagation of measure, are obviously closely related. Simply changing the direction of time in one ($t \to -t$) results in the other. In common parlance, $e^{t\mathcal{L}_f^\dagger}$ is referred to as the *forward* propagator and $e^{t\mathcal{L}_f}$ the *backward* propagator.

An important consequence of the skew symmetry of $\mathcal{L}_f$ for a divergence-free system is that the propagator $e^{t\mathcal{L}_f^\dagger}$ preserves the integral over the whole phase space, i.e.

$$\begin{aligned}\int_{\mathcal{D}} e^{t\mathcal{L}_f^\dagger}\rho(z)\mathrm{d}\omega &= \int_{\mathcal{D}} e^{-t\mathcal{L}_f}\rho(z)\mathrm{d}\omega \\ &= \int_{\mathcal{D}} \rho(\mathscr{F}_{-t}(z))\mathrm{d}\omega \\ &= \int_{\mathcal{D}} \rho(z)\mathrm{d}\omega,\end{aligned}$$

since we can view $\mathscr{F}_{-t}(z)$ as a change of variables by a volume-preserving map.

The Liouvillian of a Hamiltonian system with N_c degrees of freedom and a Hamiltonian of the form $H = \sum_{i=1}^{\mathrm{N}_c} p_i^2/(2m_i) + U(q_1, \ldots, q_{\mathrm{N}_c})$ can be written

$$\mathcal{L}_H^\dagger = -\sum_{i=1}^{\mathrm{N}_c} m_i^{-1} p_i \frac{\partial}{\partial q_i} - \sum_{i=1}^{\mathrm{N}_c} F_i \frac{\partial}{\partial p_i},$$

where $F_i = -\partial U/\partial q_i$.

5.3 The Microcanonical Probability Measure

In this section, we continue the discussion of the Hamiltonian case by introducing the microcanonical probability measure, the natural measure associated to a surface of fixed energy. The framework we have discussed so far is useful when the density is known to be a C^∞ function of phase space, but this is not always the case.